

From Hands-On Experience to Inclusive AI Education: Reflections on Gateways 2026

Dr. Oyebade Kunle Oyerinde

2026 SGX3 Rising Star | Full Professor, Clark Atlanta University | Special Assistant to the Provost for AI, Data, and Analytics | 2026 NSF NAIRR AI Education Fellow

Attending the Gateways 2026 Conference in Washington, DC, was an important opportunity to deepen my engagement with the Science Gateways community and explore how cyberinfrastructure, high-performance computing (HPC), and artificial intelligence can expand access to research, teaching, and learning. The conference reinforced my conviction that Science Gateways are not simply technical platforms; they are pathways to discovery, collaboration, and inclusive participation in advanced computing.

A central focus of my work at Clark Atlanta University is developing the graduate course *Economics and Data Science for Public Administrators* (CPAD 504). This course integrates economic theory, quantitative analysis, authentic datasets, data visualization, and responsible AI to help students address real-world public-sector and policy challenges. Through hands-on experiences with CloudBank Hub and SGX3 Gateways large language models, and publicly available datasets, I am exploring how students can use accessible computing environments to conduct analyses, visualize data, evaluate AI-generated outputs, and develop evidence-based policy recommendations. My goal is to ensure that students learn not only how to use AI and data science tools, but also how to question their outputs and apply them responsibly.

My practical experience with CloudBank Hub has helped me better understand the opportunities and challenges involved in incorporating advanced computing into university teaching. I have also used CloudBank Hub in instructional activities associated with the Atlanta University Center Data Science Initiative's pre-freshman program and in my graduate course *Research Methods and Data Analysis* at Clark Atlanta University. These experiences have strengthened my interest in making HPC and AI resources more accessible to students at historically Black colleges and universities (HBCUs), where such opportunities can support broader participation in data science and AI education.

Another significant part of my engagement with the conference was serving as a judge for six research papers. The papers addressed important issues across the Science Gateways ecosystem, including strategies for sustainable AI adoption, FAIR data compliance and gateway migration, verification-first scientific literature curation, AI integration in production gateways, prompt-augmented learning through guided codeathons, and responsible AI-enabled translational research. Reviewing these contributions gave me an

opportunity to consider how technical innovation, research rigor, practical relevance, and knowledge transfer can strengthen the broader gateways community. The paper on teaching advanced data science and gateway development through guided codeathons was particularly relevant to my interest in experiential learning and integrating AI into curriculum design.

The conference's broader lesson for me was that successful AI integration requires more than access to powerful tools. It depends on thoughtful instructional design, reliable data, responsible governance, meaningful hands-on practice, and collaboration across disciplines. The papers I reviewed and the conversations at Gateways 2026 reinforced the importance of designing systems and learning experiences that are technically sound, usable, transparent, and relevant to real-world needs.

As a Full Professor at Clark Atlanta University, Special Assistant to the Provost for AI, Data, and Analytics, and 2026 NSF NAIRR AI Education Fellow, I will carry these insights into my teaching, curriculum development, and broader AI education initiatives. I am grateful to the SGX3 community for creating opportunities to learn, contribute, and collaborate. I look forward to continuing to develop practical approaches that connect Science Gateways, HPC, and responsible AI with inclusive educational opportunities for the next generation of researchers, data scientists, and evidence-informed leaders.

My central takeaway: Science Gateways can bridge the gap between access to advanced technology and the ability to use it meaningfully. Science Gateways can also expand not only what students can do but also who gets the opportunity to participate in AI-driven discovery.